INCH-POUND

MIL-C-915/7F
29 September 1989
SUPERSEDING
MIL-C-915/7E
18 June 1985

MILITARY SPECIFICATION SHEET

CABLE, POWER ELECTRICAL, FOR OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE DSWS

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, AWG size 14, class B stranding, tin coated.
Second - Synthetic rubber insulation. Nominal wall thickness 0.073 inch. Standard identification code applied by method 3. An additional covering of clear polyamide may be employed at the manufacturer's option.
Third - Two conductors shall be cabled together with a lay not greater than 5 inches. Fillers of synthetic rubber-like compound shall be used to attain a firm, well rounded cross section. Fillers may be laid in the valleys during the cabling operation or may be extruded into the valleys separately. Fillers shall be readily separable from the insulation without damage thereto and without destroying the conductor identification.
Fourth - Binder tape over the conductor assembly, at manufacturer's option.
Fifth - Shielding braid of AWG no. 34, tin coated copper, with an angle of 30 to 40 degrees and minimum coverage of 88 percent.

AMSC N/A FSC 6145
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REQUIREMENTS - Continued

Sixth - An overall jacket (black), nominal wall thickness 0.060 inch. If needed to meet electrical tests, the jacket may comprise an inner layer of rubber insulation, such as synthetic rubber and an outer layer of rubber insulation, such as synthetic rubber and an outer layer of polychloroprene. The jacket shall be readily separable from the underlying shield, leaving the shield adequately clean for soldering. Cable surface marking required.

TABLE I. Details.

	Overall diameter	
Type and size	minimum (inch)	maximum (inch)
DSWS-4	0.770	0.800

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C, maximum.	2.78
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor (5 minutes).....	10,000
Conductor to shield (5 minutes).....	10,000
Shield to water (5 minutes).....	500
Voltage withstand - volts, dc, minimum	
Conductor to conductor (15 minutes).....	25,000
Conductor to shield (15 minutes).....	25,000
Shield to water (15 minutes).....	1,500
Insulation resistance - megohms/1000 feet, minimum.....	100
Conductor to conductor.....	200
Conductor to shield.....	200
Shield to water.....	100
Group A:	
Visual and dimensional.....	No failure
Hydrostatic (open end) - 1000 lbf/in^2, in 2 hours, in^3, maximum.....	0
Mutual capacitance - picofarad per feet (pF/ft) at 1 kilohertz (kHz).....	40 ± 10

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EXAMINATION AND TESTS - Continued

	Requirements
Group B:	
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - lbf/in^2, minimum	600
Elongation - percent, minimum	350
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1,800
Elongation - percent, minimum	300
Set - inch, maximum	3/8
Drip - at 95 ± 1°C	Zero
Group C:	
Physical (aged)	
Insulation (conductor)	
Air pressure heat	
Tensile strength - percent of unaged, minimum	70
Elongation - percent of unaged, minimum	75
Jacket (cable)	
Air pressure heat	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Hot oil immersion	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Permanence of printing (jacket) cycles, minimum	250
Cable filler removability	No failure
Shield - conformance to material, coverage and construction	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Insulation resistance per 1000 feet at room temperature from shield to water, 50 megohms, minimum after 90 days continuous submergence in water; values recorded at 1-week intervals.

Flexing endurance - 1000 cycles of bending a specimen, 4-inch maximum diameter mandrel at minus 40 ± 2°C. Upon successful completion of this test, specimen shall meet the voltage withstand test and the hydrostatic (open end) test.

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UNIT ORDERING LENGTH: 500 feet (nominal).

NOTE: Not for Air Force use.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1127-04)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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